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Source: *Population and Development Review*, Vol. 28, No. 3 (Sep., 2002), pp. 515-537

Published by: [Population Council](#)

Stable URL: <http://www.jstor.org/stable/3092839>

Accessed: 27-09-2015 16:58 UTC

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Muslim and Non-Muslim Differences in Female Autonomy and Fertility: Evidence from Four Asian Countries

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THE HYPOTHESIS THAT greater female power and autonomy produce lower fertility appears in many explanations of fertility differences and change (Dyson and Moore 1983; Cain, Khanam, and Nahar 1979; Basu 1992; Jeejeeboy 1995). This article examines the specific claim that the higher fertility of Muslims (compared to non-Muslims) can be traced to the lower level of power and autonomy (hereafter autonomy) afforded Muslim women.¹ This lower autonomy reflects more rigid forms of patriarchy that characterize Muslim communities. Key elements of this argument “inhere at the systemic, institutional, aggregate level” and thus require analysis at the community level (Smith 1989: 173–174; see also Balk 1994; Jejeebhoy and Sathar 2001). From a set of surveys of more than 50 communities in four Asian countries, we identify 14 Muslim/non-Muslim pairwise community comparisons—that is, pairs of communities that share many characteristics but differ in that one is predominately Muslim and the other predominately non-Muslim. Although not representative of any country or region, these pairs of communities reflect substantial diversity of social and economic setting. Across most pairs of settings, we show that Muslims have more children, are more likely to want another child, and, if they want no more children, are consistently less likely to be using contraception. We test whether these differences parallel differences in married women’s autonomy.

We use data collected for the purpose of investigating women's power, autonomy, and fertility: the Survey on the Status of Women and Fertility (SWAF). These data, collected in 1993 and 1994, include a number of potentially relevant dimensions of women's autonomy: freedom of movement, economic autonomy, and exposure to intimidation/coercion. Direct measurement of multiple dimensions of autonomy avoids many of the criticisms of prior research on "women's status" and fertility (as discussed in Mason 1984, 1986, 1987).

Religion, women's autonomy, and reproductive behavior

Can one reasonably argue that the Islamic religion exerts consistent effects on demographic behavior? Caldwell proposes such an overarching effect of Islam, rooted in patriarchy. To the extent that Islam is more strongly patriarchal than other religions, demographic behavior could be influenced (also see Kirk 1968; Balk 1994). With regard to mortality differentials observed between Muslims and non-Muslims in a range of countries presented in the *World Development Report 1984*, Caldwell (1986: 175) writes,

These mortality differences are not necessarily inherent in religions, nor immutable.... They are important at present, however. The central aspect of the relationship between Islam and mortality levels is undoubtedly the separate and distinctive position of women operating partly through their access to education but also in many other ways.

Although Caldwell's argument is framed to address mortality experiences, it can be extended to fertility. Constraints on behavior, stemming from strongly patriarchal systems, could result in 1) a desire for more children, and, 2) given that no more children are desired, lower rates of contraceptive use. Such results could be produced by two related processes:

First, associations could operate at the aggregate level. Accordingly, Muslim communities would be characterized by less autonomy for women and greater pronatalism. Within communities, variance in autonomy would be modest and would be inconsequential for fertility outcomes.

Second, powerful individual-level associations could mediate the aggregate association between religion, women's autonomy, and fertility. Specifically, women with greater autonomy might desire fewer children and be more likely to use contraception. Religion or ethnicity might operate by shifting the distributions of autonomy among women in Muslim versus non-Muslim settings.

Consistent with both processes, Cain, Khanam, and Nahar (1979: 432) described linkages between religion, women's autonomy, and the demand for children in their study of class and patriarchy in Bangladesh:

Male dominance is...supported by interlocking and reinforcing elements of kinship, political, and religious systems. Powerful norms of female seclusion extend to labor markets, severely limiting women's opportunities for independent income generation.... The risk and insecurity that patriarchy imposes on women represent a powerful systemic incentive for high fertility.

In other words, patriarchal systems can increase the demand for children because they usually limit women's nonfamilial opportunities for social status and economic support. Where women's opportunities outside the home are severely constrained, their survival strategies focus inward on family and children. Bearing children, especially sons, solidifies a woman's position in her husband's or in-laws' house; "respect, protection, or claim on family resources" can depend on high fertility (Mason and Taj 1987: 615). Sons may also be in a position to provide for their mother's old-age security or for their care in the case of widowhood (Cain, Khanam, and Nahar 1979: 433).

Finally, hypotheses linking women's autonomy to contraceptive use have long been advanced in the literature. For example, where group norms and practices limit women's mobility and their contact with nonfamily members, women's exposure to novel ideas or technological innovations, including contraceptives, may be constrained (Cleland and Wilson 1987). Opposition from husbands can also be a barrier to contraceptive use in patriarchal settings. Acquiring contraceptives may be especially difficult in situations where limitations are placed on women's freedom of movement and access to economic resources.

A competing explanation argues that Muslim and non-Muslim fertility differences result from different levels of economic development or different socioeconomic levels. Jeffery and Jeffery (1997: 225; also see Weeks 1988) suggest that differences in fertility rates between Hindus and Muslims in India are generally the result of underlying differences in "region, residence, class and schooling," rather than differences in religious orientation or in the autonomy of women. Our data will allow a test of this competing hypothesis.

A third explanation attributes reproductive differences to religious doctrinal differences, especially ones relating to the acceptability of contraception and abortion. Although generally perceived to be pronatalist, the Qur'an does not contain straightforward injunctions against the use of contraception. Obermeyer (1994a) concludes that, on the basis of religious dogma, it would be difficult to substantiate claims that Muslims are more pronatalist or less approving of contraception than are other groups. In keeping with this argument, Muslims do not consistently give reasons for nonuse of contraception that suggest they believe it is "against their religion."² Omran's (1992: 102-111) review of key writings indicates that Allah directs adherents to have children and that Allah has granted provision for all His creatures, including any children that are conceived. However, marriage and

parenthood are not synonymous with unrestricted childbearing; and the Qur'an stresses the care and well-being of children over the bearing of large numbers of children (ibid.). Most important, supporters of family planning cite Islamic jurisprudence that interprets silence on the issue of contraception neither as an error of omission nor as a reflection of an absence of population pressures in past times. Rather, the *Lawgiver* is all-knowing and the Qur'an is for all-time. Further, withdrawal was practiced as a contraceptive measure at the time the Qur'an was revealed, and the Prophet offered no prohibition. The majority of Islamic theologians consider this to be unequivocal evidence that withdrawal and other nonpermanent methods of birth control are acceptable. Sterilization is more problematic in Islamic jurisprudence (see Omran 1992: Chapters 10 and 11).³

In contrast to these broad explanations that might produce a general pattern of higher Muslim fertility, a fourth explanation is particularistic and interactive. The particular insecurities or constraints a group faces can produce fertility responses, either antinatalist or pronatalist ones. A substantial literature on such differences exists (e.g., Goldscheider and Uhlenberg 1969; Kennedy 1973; Johnson 1979; Johnson and Nishida 1980), but we maintain a focus on Asia and Muslims. Specifically, in a comparative study of a Hindu (Jat) and a Muslim (Sheikh) community in Uttar Pradesh, India, Jeffery and Jeffery (1997) found that Muslim/non-Muslim fertility differentials remained after socioeconomic controls. Moreover, Jeffery and Jeffery (1997: 161) found few differences between Jats and Sheikhs in the degree of "freedom of action" that women enjoy. Thus to explain fertility differentials, they adopt a context-specific (as opposed to a global) interpretation. They argue that the higher "fertility of the Sheikhs can be understood much more in terms of their marginalized position as part of a minority group within Indian society, rather than as a response to any 'essential' feature of Islam" (ibid.: 229).⁴ Identity issues and ideologies that justify or celebrate differences can arise from group competition and conflict. As Obermeyer (1994a: 51) notes, "Like other religious doctrines, Islam has been used to legitimate conflicting positions on gender and reproductive choice" (see also Omran 1992: 85–112). Likewise, in Thailand, Knodel et al. (1999) document substantial fertility differences between Muslims and Buddhists in some areas but moderate differences elsewhere. They argue that explanation requires attention to local historical, political, and cultural context.

We examine the fit of these various explanations to Muslim/non-Muslim differences across a range of settings. Are fertility differences pervasive, suggesting general or even global processes? Or are they idiosyncratic, suggesting context-specific explanations? If global explanations seem warranted, are differences in autonomy a plausible explanation? Or can socioeconomic disadvantage better account for differences?

Data and measurement

As stated earlier, the communities in our study are not necessarily representative of particular countries or regions. Even with random sampling, there is little chance that a handful of communities would be representative of a national population. Instead, the communities selected have a set of characteristics, primarily Muslim versus non-Muslim, conforming to our research design. The Muslim communities in the study are by no means homogenous, either within or between countries, and they are not uniformly inferior to their non-Muslim neighbors in measures of socioeconomic status. Our goal is not to give a precise measurement of Muslim versus non-Muslim differences in fertility expectations and contraceptive use in these countries, but rather to ascertain that such differences are (or are not) ubiquitous in a range of settings, and to ask whether such differences (or lack thereof) can be attributed to measurable differences in women's status as between Muslim and non-Muslim communities. We acknowledge below several weaknesses of our design, but its strengths more than compensate for them. A useful referent would be the common strategy of choosing a single set of contrasting villages (e.g., Jeffery and Jeffery 1997; Dharmalingam and Morgan 1996; Morgan and Niraula 1995). Single comparisons make generalization difficult; more numerous comparisons, in diverse settings, allow broader generalization.

We use a subset of data from a five-country study of the relationship between fertility and the status of women in communities in South and Southeast Asia (SWAF).⁵ The countries in the original study were India, Malaysia, Pakistan, the Philippines, and Thailand. The communities in this investigation are restricted to Muslim communities and their logical comparisons, namely non-Muslim communities that share characteristics, including geographical proximity. Data from India are presented separately for the states in which they were collected (Uttar Pradesh and Tamil Nadu), owing to the well-known differences in fertility and family structure associated with north and south India (Dyson and Moore 1983). We omit data from Pakistan (in the absence of non-Muslim communities for comparison; but see Jejeebhoy and Sathar 2001) and from regions of Thailand other than the south (because only the south provides Muslim communities for comparison). A comparative organization of communities is shown in Table 1 and in the row structure of subsequent tables. Table 2 presents information on the communities' social and economic organization.

The research settings

Within each country, the organizing sampling unit was the community, variously defined but having in common fixed geographic boundaries and homogeneity along relevant religious and/or ethnic dimensions. Primary sampling

TABLE 1 Controlled comparisons of Muslim and non-Muslim communities:
Status of Women and Fertility data

Country and communities	Matched characteristics	Religious or ethnic contrast
India: Uttar Pradesh		
1. Pratapgarh	Less developed subdistricts	Muslim vs. Hindu
2. Meerut	More developed subdistricts	Muslim vs. Hindu
India: Tamil Nadu		
3. Ramnathpuram	Less developed subdistricts	Muslim vs. Hindu
4. Coimbatore	More developed subdistricts	Muslim vs. Hindu
Malaysia		
5. Kelantan & Selangor	Rural	Muslim/Malay vs. Hindu/Indian
6. Kelantan & Selangor	Rural	Muslim/Malay vs. Chinese
7. Kelantan & Selangor	Rural	Muslim/Malay market women vs. Hindu/Indian
8. Kelantan & Selangor	Rural	Muslim/Malay market women vs. Chinese
9. Selangor	Urban	Muslim/Malay vs. Hindu/Indian
10. Selangor	Urban	Muslim/Malay vs. Chinese
Thailand		
11. Pattani/Trang	Rural, south	Muslim/Malay vs. Thai Buddhist
12. Songkhla/Nakhon Si Thammarat	Rural, south	Muslim/Thai vs. Thai Buddhist
Philippines		
13. Zamboanga/Oriental Mindoro	Rural, similar subsistence	Muslim vs. Christian
14. Zamboanga/La Union, Camarines Sur	Rural	Muslim vs. Christian

units (PSUs) were variously provinces (Thailand, Philippines), states (Malaysia), or districts within states (India). Except in Thailand, these PSUs were not selected at random, but rather purposively along dimensions of interest in the study of the status of women (e.g., economic well-being, types of agricultural production, presence of Muslim communities). Subunits within PSUs were randomly selected and were variously municipalities and (below them) *barangays* (Philippines), districts, *tambols*, and villages (Thailand), districts, towns, and neighborhoods (Malaysia), and *talukas* (subdistricts) and villages (India). There was some aggregation of the lowest-level subunits for purposes of defining a community, for example villages in India and *barangays* in certain municipalities in the Philippines. Within subunits (or communities), sampling of women was random with occasional quota sampling to obtain a community-level sample of some fixed size, for example in the Philippines.

In India, several sites contained villages in which both Hindus and Muslims lived but where, despite their physical proximity, the groups ap-

TABLE 2 Social and economic characteristics of Muslim and non-Muslim populations (means and percentages)

Country/ community	Non-Muslim comparison	Married to a relative (percent)		Wife's age (years)		Wife's age at first marriage (years)		Wife's education (years)		Husband's education (years)		Household possessions index (0-10)		Number of observations	
		Muslim	Non- Muslim	Muslim	Non- Muslim	Muslim	Non- Muslim	Muslim	Non- Muslim	Muslim	Non- Muslim	Muslim	Non- Muslim	Muslim	Non- Muslim
India: Uttar Pradesh															
1. Pratapgarh	Hindu	33.8	2.3	28.2	27.1	15.4	12.3	2.0	2.6	5.3	8.1	2.8	2.0	213	214
2. Meerut	Hindu	16.4	1.8	26.7	28.1	15.3	16.3	0.0	3.1	3.4	8.3	1.1	3.6	207	223
India: Tamil Nadu															
3. Ramanathapuram	Hindu	80.2	71.0	28.9	28.9	17.1	17.9	3.0	1.9	5.2	3.8	1.8	1.4	339	210
4. Coimbatore	Hindu	35.7	41.8	29.1	29.1	17.9	18.6	4.2	4.1	5.5	5.4	2.4	2.5	207	225
Malaysia															
5. Rural Malays	Hindu	23.7	59.7	30.0	30.7	19.8	21.0	8.5	5.3	9.0	6.5	7.3	6.7	101	191
6. Rural Malays	Chinese	9.2	9.2	30.7	30.7	21.4	21.4	5.5	5.5	5.7	5.7	7.9	7.9	196	196
7. Market Malays	Hindu	14.6	59.7	30.5	30.7	20.1	21.0	9.7	5.3	9.8	6.5	7.9	6.7	205	191
8. Market Malays	Chinese	9.2	9.2	30.7	30.7	21.4	21.4	5.5	5.5	5.7	5.7	7.9	7.9	196	196
9. Urban Malays	Hindu	23.3	44.0	31.4	30.8	20.2	20.7	9.1	6.7	8.2	7.6	7.7	7.0	210	175
10. Urban Malays	Chinese	3.3	3.3	31.2	31.2	22.3	22.3	7.9	7.9	8.4	8.4	8.4	8.4	183	183
Thailand															
11. Malay-speakers	Buddhist	30.8	1.8	29.5	29.6	16.9	20.7	5.2	7.2	4.4	7.5	4.3	6.0	81	113
12. Thai-speakers	Buddhist	41.9	5.3	30.1	29.3	18.8	19.2	5.8	5.6	5.7	6.2	6.2	4.7	90	112
Philippines															
13. Zamboanga	Christian	37.5	3.5	28.7	28.9	18.7	20.2	4.5	7.1	5.3	6.6	2.8	2.2	200	200
14. Zamboanga	Christian	4.0	4.0	29.9	29.9	20.7	20.7	8.2	8.2	7.8	7.8	2.8	2.8	200	400

peared to form distinct social communities. There are four Muslim/non-Muslim contrasts in India, two in the north (Uttar Pradesh) and two in the south (Tamil Nadu). Within each state, two districts were selected, one relatively more developed and one relatively less developed, and communities were sampled from within subdistricts of each district, again with a socioeconomic development differential. Notwithstanding this attempt at stratification by socioeconomic status, there is one community—Muslims in a subdistrict of Pratapgarh, in Uttar Pradesh—where socioeconomic differences between neighboring Muslim villages exceed their aggregate differences relative to their Hindu neighbors.

The Malaysian communities are defined by religio-ethnic category and urban versus rural residence. The religio-ethnic categories are Malays, who are Muslim; Indians, Hindus only; and Chinese, who can be variously categorized with regard to religious affiliation.⁶ In Malaysia, unlike India, the distinctions among these groups are as much ethnic as they are religious (Leete 1996: Chapter 1). The urban samples come from a port city in the state of Selangor. The rural Indian and Chinese communities are found in the outskirts of this city; the rural Malay community comes from the state of Kelantan, near the border with Thailand. Also from this area is a community of Kelantanese market women, rural by origin but trading daily in an urban area. As can be seen in Table 2, these Malaysian communities are strongly differentiated by the extent of consanguineous marriage, which is most common among Hindus (with origins primarily in South India), less so among Malay Muslims, and least so among Chinese. There are comparatively few socioeconomic differences among the urban Malaysian groups. The comparison between rural Kelantanese Malays and their rural Chinese and Hindu “controls” (who are in Selangor) suffers from the geographic and provincial separation of the Malays from the other two groups; it has the compensating virtue that the rural Malays are advantaged in terms of education, thus giving them a theoretical “leg up” in prospects for fertility control.

Muslim communities in Thailand are found primarily in the country’s south—in our study, near the border with Malaysia.⁷ These Thai Muslim communities are ethnically Malay; they are best distinguished from their ethnic Thai neighbors by their high proportions married to relatives. Otherwise, as shown in Table 2, the Muslim community in Pattani has lower levels of education than the Muslim community in Songkhla, which is educationally similar to (and perhaps wealthier than) one of the two rural Thai Buddhist communities.

The Philippine data included rural Muslims from the country’s south but no Christians from that area. Thus, the Philippine data provide the weakest opportunity for a true controlled comparison. Given these data, Muslim and non-Muslim differences are not separable from regional differences. With this weakness in mind, we compare southern Philippine Muslims (from

Zamboanga) with a rural population that has a similar economic base (Mindoro, in the central Philippines), and with two other rural communities aggregated (one from the north, from La Union, and one from the mid-central region, from Camarines Sur). The Philippine communities are uniformly poor, at least with regard to household possessions. Educational levels are, however, comparatively high but are lowest in the Muslim communities.

Measurement of autonomy

We consider three aspects of wives' power and autonomy: 1) their freedom of physical movement, that is, whether wives need permission before going to various places; 2) their say in economic decisions; and 3) the interpersonal control exercised by husbands, both psychological and physical. While not exhaustive, these measures draw from many aspects of power and autonomy that wives confront daily.

The first aspect of women's autonomy—their freedom of movement—is likely to be crucial if women are to translate a desire to end childbearing into effective contraceptive use. Freedom of movement is a signal feature of autonomy, namely the ability "to obtain information and to use it as the basis for making decisions about one's private concerns and those of one's intimates" (Dyson and Moore 1983: 45). Freedom of movement was measured by asking whether a woman must get permission before visiting various places: the community center, the local market, a health center or clinic, the fields around the village, and the homes of friends and relatives. Wives' freedom of movement score is increased by 1.0 for each place she can go without permission. In the Thailand data, only a general question was asked: whether there was any place the woman cannot go without getting permission from her husband or other family members. For the Thai communities we substitute this dichotomous measure for the index described above.⁸

Women's say in economic decisions is measured through a seven-point scale created by summing responses to six dichotomous items. A woman was asked whether or not she 1) had any say in household decisions about major purchases; 2) had any say in decisions about whether she should work outside the home; 3) had the sole or greatest say in major purchasing decisions; 4) had the sole or greatest say in decisions about her employment; 5) felt free to buy a dress or sari without getting permission from other household members; and 6) felt free to buy a small item of jewelry without getting permission from other household members.⁹ Previous analysis of this scale has shown that it does not simply reflect the household's income level (Mason, Smith, and Morgan 1998).

Third, the interpersonal controls exercised by the husband are measured through two items: the first asks a woman whether she is afraid to disagree with her husband for fear he will become angry with her. The sec-

ond item asks whether the husband ever hits or beats her. We combine these dichotomous items into a noncoercion scale with range 0–2 (i.e., 2 if not afraid to disagree and if husband has never hit or beat her).¹⁰

Ghuman, Lee, and Smith (2000) have examined the measurement properties of these items by taking advantage of parallel reports from wives and husbands. Their analysis suggests that the items have similar meanings and metrics within countries but not between countries. Such a result provides an additional rationale for the paired within-country contrasts on which our analysis focuses.

Findings

In a review of dominant fertility theories Smith (1989: 171) writes, “The most important causal factors in these theories—typically social or cultural institutions, occasionally emergent properties of the collective behavior of individuals—are properly conceptualized at the macro level.” Our analysis follows Smith by adopting a comparative research design that features variation at the macro level. Smith also called for complementary analysis at the individual level, in recognition of the processes through which systemic properties are translated into individual behaviors. As such, our analysis begins with 14 community comparisons, focusing primarily on pairwise differences in community means with regard to measures of women’s autonomy and fertility. Subsequently, we test the association between religion, women’s autonomy, and the outcomes within matched community pairs to observe effects operating at the individual level. Our findings are organized around four general questions.

Across matched Muslim and non-Muslim communities,

1) Are there consistent differences in the fertility behavior of Muslims and non-Muslims?

2) Are there consistent differences in the levels of autonomy afforded women between Muslim and non-Muslim groups?

3) Are larger aggregate differences in autonomy found in the same settings as larger differences in fertility outcomes (i.e., a comparison of differences between differences)?

Within samples of matched communities,

4) Are individual-level differences in autonomy associated with fertility outcomes, and does their inclusion in a multi-level model attenuate the effects of religion on fertility outcomes?

With respect to the first question, Table 3 shows the mean number of children ever born to women married ten or more years,¹¹ the proportion of all wives who want more children, and among those who do not want additional children the proportion not using contraception (i.e., those at a higher risk of an unwanted pregnancy). All estimates are standardized for age and marital duration or parity. Focusing first on column 1 (i.e., mean

TABLE 3 Mean number of children ever born, percent of women wanting more children, and percent of women not using contraception among those wanting no more children, for contrasting pairs of Muslim and non-Muslim communities

Country/ community	Non-Muslim comparison	Mean number of children ever born ^a		Percent wanting more children ^b		Percent not using contraception among those not wanting more children ^c	
		(1)	(2)	(3)	(4)	(5)	(6)
	Prediction:	Muslim > Muslim	Non-Muslim > Muslim	Muslim > Muslim	Non-Muslim > Muslim	Muslim > Muslim	Non-Muslim > Muslim
India: Uttar Pradesh							
1. Pratapgarh	Hindu	5.42	4.50	60	45	76	55
2. Meerut	Hindu	5.11	4.46	48	28	60	29
India: Tamil Nadu							
3. Ramnathpuram	Hindu	4.48	4.53	36	37	62	40
4. Coimbatore	Hindu	3.54	2.49	25	25	25	21
Malaysia							
5. Rural Malays	Hindu	4.96	4.23	87	42	57 ^d	26
6. Rural Malays	Chinese	4.96	4.35	87	48	57 ^d	22
7. Market Malays	Hindu	3.99	4.23	89	42	60	26
8. Market Malays	Chinese	3.99	4.35	89	48	60	22
9. Urban Malays	Hindu	4.61	3.84	69	41	48	18
10. Urban Malays	Chinese	4.61	3.82	64	55	48	12
Thailand							
11. Malay-speakers	Buddhist	3.77	3.08	74	42	79	15
12. Thai-speakers	Buddhist	2.97	3.15	35	40	26	6
Philippines							
13. Zamboanga	Christian	5.32	5.13	77	39	59	46
14. Zamboanga	Christian	5.32	4.55	77	36	59	40
Correct prediction and difference is significant ^e		8 out of 14 contrasts		11 out of 14 contrasts		12 out of 14 contrasts	

^aIncludes only women married ten or more years; standardized for age.

^bIncludes women currently pregnant, who were asked about additional children; standardized for age and parity.

^cExcludes women currently pregnant; standardized for age and parity.

^dNot standardized—sample too small.

^eCounts only statistically significant differences ($p < .05$) in the predicted direction.

differences in children ever born), we see substantially higher fertility among Muslims than Hindus in Uttar Pradesh, regardless of the socioeconomic level of the community. Fertility is substantially lower in Tamil Nadu for both Muslims and Hindus. Also note that in Tamil Nadu a sharp difference is observed between Muslims and non-Muslims only in contrast 4, that is, when communities from the more developed district or subdistrict are contrasted. Thus, these data show higher Muslim fertility in three of the four Indian settings.

In addition to these differences in parity, column 2 shows that Muslims compared to non-Muslims are more likely to report wanting more children in the north Indian contrasts (i.e., communities 1 and 2) but not in the south (communities 3 and 4). Finally, among women who want no more children, Muslims are more likely to be at high risk of an unwanted pregnancy (i.e., not wanting more children and not using contraception) in all four Indian contrasts. Female sterilization (surgical contraception) is a major form of birth control in all communities studied, both in India and elsewhere (results not shown here). Moreover, differences in the proportion sterilized compared to the proportion using other modern methods account for much of the observed differences in contraceptive use between Muslim and non-Muslim groups.

Results for other countries are generally similar to those for India, that is, Muslims are more likely to want more children, but there are exceptions. In Malaysia, Malays have the expected differences in desire for more children in both rural and urban areas over both Indians and Chinese. The Malay market women (with substantial freedom of movement, as we will document) have fertility levels that approximate those of Indian and Chinese women. On the subsequent measures, Malays are more likely to report a desire for more children, and, when they do not want more children, they are less likely to use contraception.

Results vary substantially across the two Thai contrasts. The first Thai contrast provides sharp differences in the expected direction (i.e., the Muslim community shows a desire for more children). The second contrast shows higher fertility among non-Muslims and a roughly equal proportion reporting a desire for more children. We do not understand why results differ so greatly, but Table 2 shows that the two sampled Muslim settings differ on a variety of dimensions. As in all previous contrasts, Muslim wives who want no more children are less likely to be using contraception. Finally, all Philippine contrasts conform to predictions of greater Muslim desires for more children.

Overall, then, these community contrasts show strikingly different levels of fertility, desires regarding additional children, and contraceptive use. The last row of Table 3 shows that 8 out of 14, 11 out of 14, and 12 out of 14 differences are statistically significant and conform to expectations that Muslims would indicate a desire for more children. Given that clear differences exist, these data provide an appropriate setting for exploring the reasons for observed differences.

Do Muslim and non-Muslim communities show parallel differences in levels of autonomy? Such differences would be consistent with arguments that Muslim communities are more patriarchal. Table 4 shows mean values for the freedom of movement, economic power, and noncoercion indexes for Muslims and non-Muslims in each of the 14 community comparisons.

TABLE 4 Mean scores for indexes of freedom of movement, economic power, and noncoercion, for contrasting pairs of Muslim and non-Muslim communities

Country/ community	Non-Muslim comparison	Freedom of movement index ^a (1)		Economic power index ^a (2)		Noncoercion index ^a (3)	
		Prediction:	Non- Muslim < Muslim	Muslim < Muslim	Non- Muslim < Muslim	Muslim < Muslim	Non- Muslim < Muslim
India: Uttar Pradesh							
1. Pratapgarh	Hindu	0.61	0.72	1.30	1.18	1.37	1.18
2. Meerut	Hindu	0.28	0.29	1.24	1.34	1.02	1.08
India: Tamil Nadu							
3. Ramnathpuram	Hindu	2.59	2.91	1.72	2.04	1.67	1.58
4. Coimbatore	Hindu	2.06	2.85	1.89	2.32	1.40	1.60
Malaysia							
5. Rural Malays	Hindu	2.00	1.69	1.24 ^b	0.80 ^b	0.54 ^c	0.52 ^c
6. Rural Malays	Chinese	2.09	4.34	1.24 ^b	1.51 ^b	0.54 ^c	0.75 ^c
7. Market Malays	Hindu	2.80	1.69	2.71	2.61	0.58 ^c	0.52 ^c
8. Market Malays	Chinese	2.80	4.34	2.71	3.13	0.58 ^c	0.75 ^c
9. Urban Malays	Hindu	2.26	2.09	2.69	2.35	0.56 ^c	0.50 ^c
10. Urban Malays	Chinese	2.26	4.76	2.69	3.63	0.56 ^c	0.84 ^c
Thailand							
11. Malay-speakers	Buddhist	0.13 ^d	0.49 ^d	2.64	3.71	1.27	1.11
12. Thai-speakers	Buddhist	0.27 ^d	0.44 ^d	4.03	3.84	1.12	1.03
Philippines							
13. Zamboanga	Christian	0.29	3.00	3.65	3.66	1.10	1.35
14. Zamboanga	Christian	0.29	3.18	3.65	3.41	1.10	1.56
Correct prediction and difference is significant ^c		9 out of 14 contrasts		4 out of 14 contrasts		6 out of 14 contrasts	

^aEstimates for paired Muslim and non-Muslim groups are based on a common age distribution, i.e., standardized for age.

^bBased on two items only: purchase of sari or dress, and purchase of jewelry.

^cBased on single item only: fear of disagreeing with husband.

^dBased on single item only: not free to go everywhere one chooses.

^eCounts only statistically significant differences ($p < .05$) in the predicted direction.

Looking first at the Indian contrasts, we see large differences between Uttar Pradesh and Tamil Nadu on freedom of movement; as expected women in the south (i.e., Tamil Nadu) exhibit greater freedom of movement (Dyson and Moore 1983; Basu 1992; Jejeebhoy and Sathar 2001). But within regions, differences are modest; nevertheless in three of the four contrasts Muslims have lower freedom of movement scores (in the fourth the difference is trivial). In Malaysia, there are large differences between Muslims and Chinese, with Chinese women exhibiting very high scores for autono-

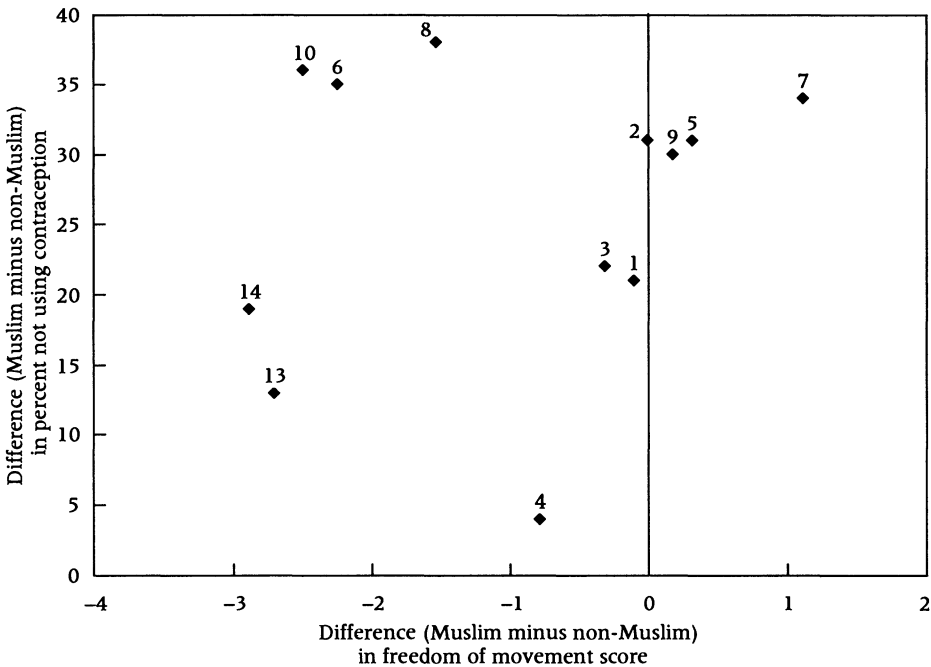
mous movement. But in all three cases, Malays (Muslims) have higher values than Indians (Hindus) although these differences are not large. Further, the rural women who are engaged in market work, which requires substantial autonomous movement, show the expected higher values on the freedom of movement index compared with other rural Malay women. In Thailand and the Philippines, differences are large and consistent: Muslims have less freedom of movement than non-Muslims (Thai Buddhists and Christian Filipinos). In sum, Muslims have reduced freedom of movement compared to non-Muslims, but this finding is not without exceptions: only 9 of 14 contrasts show that Muslims have significantly less freedom of movement (see last row of Table 4).

Examination of differences on the other two measures shows less support for the claim of less autonomy among Muslim women. Only 4 out of 14 contrasts show Muslims with significantly less economic power; in only 6 out of 14 contrasts is the difference on the noncoercion index significant and in the predicted direction. These weak links between the Muslim/non-Muslim contrasts and measures of women's autonomy are consistent with additional analyses reported elsewhere (Mason, Smith, and Morgan 1998).

We now ask the third and central question: Are larger religious/ethnic, aggregate differences in autonomy found in the same settings as larger differences in the fertility-related measures (i.e., a comparison of differences between differences)? A "yes" answer implies an aggregate-level association. In 9 out of 14 cases, differences calculated from the freedom of movement index (column 1 of Table 4) and differences in desire for additional children (column 2 of Table 3) are both in the predicted direction. That is, Muslims have lower freedom of movement and are more likely to desire additional children compared to their matched non-Muslim community. Similarly, differences in the economic power index and differences in being at risk of an unwanted pregnancy both conform to predictions in 8 out of 14 cases. Only limited evidence of an aggregate-level association can be found in these comparisons and in others not presented here. Figure 1 is illustrative and focuses on the autonomy and fertility measures that showed the greatest Muslim/non-Muslim difference. Using standardized means from Tables 3 and 4, Figure 1 shows a scatterplot of Muslim and non-Muslim differences (Muslim minus non-Muslim) on the freedom of movement scale by differences in the proportion at risk of an unwanted pregnancy. Lower autonomy among Muslims than non-Muslims (indicated by negative values on the x-axis) should be associated with larger differences in the proportions at risk of an unwanted pregnancy (indicated by larger positive values on the y-axis), that is, a negative association. No such association is visible.

Inspection of particular comparisons underlines this finding. The contrasting communities with the largest and smallest differences in the proportion at risk of unwanted pregnancy, on the vertical scale (respectively, community 8. Malaysia: Rural Malay market women vs. Chinese; and com-

FIGURE 1 “Differences between differences”: Contraception and freedom of movement



NOTE: Numbered points represent communities listed in column 1 of Table 1. Communities 11 and 12 are not shown because the Thai data included only a single item measuring freedom of movement, thus making estimates for 11 and 12 not comparable to others plotted here.

munity 4. India, Tamil Nadu: Muslim/Hindu) show only modest differences on the autonomy measure (the horizontal scale). The Philippine communities (13 and 14) show the greatest Muslim deficit of freedom of movement (i.e., values of nearly -3) but have only a modestly higher risk of an unwanted pregnancy compared to Christians (approximately 16 percentage points higher). Two of the Malaysian controlled comparisons, community 5 (rural/Hindu) and community 7 (rural market Malays/Hindu), provide unusual settings where Muslim freedom of movement is greater than in the control group (in these cases a Hindu community). To be consistent with expectations, these Muslim settings should not show an excess proportion at risk of an unwanted pregnancy compared to groups that have less autonomy. Yet the excess proportion at risk, shown as the distance above the horizontal axis, is substantial. Moreover, the within-country comparison between communities 5 and 7 shows that rural Malay market women have a higher autonomy score and a higher excess proportion at risk of unwanted pregnancy. Again the hypothesis that greater female autonomy produces lower fertility would lead to the opposite result. Comparisons between communities 9 and 10 are urban Malaysian contrasts of Muslims with Hindus and Chinese, respectively. Here the Muslims show substantial defi-

cits in freedom of movement compared to Chinese but rough parity with Hindus (i.e., communities 9 and 10 differ substantially on the horizontal scale). In comparison, Muslims show approximately equal excesses in the proportion at risk of an unwanted pregnancy vis-à-vis both comparison groups (i.e., they are comparable on the vertical scale).

The Indian data provide the strongest design for testing our hypothesis in the sense of having better-matched controlled comparisons between Muslim and non-Muslim groups. However, Muslim deficits (compared to Hindus) on the freedom of movement scale are modest compared to some others observed. The variation observed across these four contrasts (communities 1 through 4) is positively associated with the proportion at risk of an unwanted pregnancy—a pattern counter to our hypothesis. In sum, there is little evidence in this figure of the predicted, negative association. The correlation calculated from these data is .15—not in the predicted direction and of very modest magnitude.

The shaded area of Table 5 shows similar correlation coefficients measuring the associations between differences on the full range of autonomy measures and differences on the fertility variables. As described above, negative correlations conform to expectations (that Muslims' lower autonomy is associated with more pronatalist outcomes). Most correlations are modest in magnitude, and 4 out of 9 comparisons (in the shaded area) do not

TABLE 5 Correlation of (Muslim/non-Muslim) differences on selected variables

	Freedom of movement (1)	Economic power index (2)	Noncoercion index (3)	Children ever born (4)	Want more children (5)	Among those not wanting more children, per- cent not using contraception (6)
(1)	1.00 (14)	.32 (10)	.85 [^] (6)	-.11 (12)	-.06 (12)	.15 (12)
(2)		1.00 (14)	-.36 (8)	-.15 (12)	.15 (12)	-.55 [^] (12)
(3)			1.00 (14)	.22 (8)	-.49 (8)	.48 (8)
(4)				1.00 (14)	-.09 (14)	-.07 (14)
(5)					1.00 (14)	.38 (14)
(6)						1.00 (14)

[^] Significant at .10 level

NOTE: Numbers in parentheses represent number of observations (i.e., matched communities listed in Table 1).

show the predicted negative sign. Only one correlation is significant at the .10 level—not impressive given that nine correlations are estimated. Overall, across multiple measures of autonomy and fertility, we find little evidence of an aggregate-level negative association between religious/ethnic differences in autonomy and differences on the fertility variables.

Our final question is whether aggregate-level associations result “via their effects on the conditional distributions of individual-level characteristics associated with fertility” or whether “community-level variables...account for the observed pattern of context effects” (Smith 1989: 181). This question is less interesting given the results above, namely that aggregate-level associations are, at best, modest. Our results suggest that differences between Muslims and non-Muslims in autonomy play only a limited role in explaining religious or ethnic fertility differences. Table 6 provides further evidence through a series of logistic regression models. We chose autonomy and fertility variables that were most strongly associated with religion or ethnicity in Tables 3 and 4: freedom of movement and being at risk of an unwanted pregnancy. (Results are very similar if we substitute other autonomy measures for freedom of movement or if we substitute desire for more children for contraceptive use.) For each of the 14 contrasting communities, we estimate the effect of religion or ethnicity net of age and parity (column 1). We then introduce the measure of the woman’s freedom of movement (column 2). The contrast of the two coefficients (i.e., 2.11 and 2.15, for the first community contrast) provides evidence on the role of freedom of movement as an intervening mechanism (producing a greater likelihood of being at higher risk of an unwanted pregnancy). In the case of the first community contrast (row 1), there is no attenuation of the religious/ethnic effect. Muslims are slightly more than twice as likely as Hindus to be at a higher risk of an unwanted pregnancy regardless of scores on the freedom of movement index. Columns 3 and 4 repeat the exercise with additional control variables for family income, an index of household possessions, and a measure of the wife’s educational attainment. Net of additional controls, the estimated religious/ethnic effect is larger (i.e., 2.56) than in column 1; but, as before, no attenuation of the effect accompanies the control on freedom of movement (comparing estimates in columns 3 and 4). Parallel results are obtained for all 14 contrasts.

Columns 5 and 6 show the effect of freedom of movement estimated in models given in columns 2 and 4. Our hypothesis is that these coefficients would be less than 1.0 (i.e., higher values on the freedom of movement index would be associated with a reduced likelihood of being at risk of an unwanted pregnancy). In neither model (column 5 or 6) and in none of the 14 contrasting communities is there a statistically significant effect of freedom of movement. Moreover, effects are as likely to be greater than 1.0 as less than this value. Thus, no strong or consistent evidence suggests that

TABLE 6 Estimated effects of an individual's religion or ethnicity and freedom of movement on being at risk of an unwanted pregnancy: Status of Women and Fertility data, Muslim/non-Muslim community contrasts

Country/ community	Non- Muslim com- parison	Age, parity controls ^a (1)	Age, parity, freedom of movement ^b (2)	Age, parity, SES controls ^c (3)	Age, parity, SES, freedom of movement ^b (4)	Age, parity, freedom of movement ^b (5)	Age, parity SES, freedom of movement ^b (6)
India: Uttar Pradesh							
1. Pratapgarh	Hindu	2.11*	2.15*	2.56*	2.57*	1.06	.98
2. Meerut	Hindu	3.55**	3.65**	2.59**	2.67**	.85	.85
India: Tamil Nadu							
3. Ramnathpuram	Hindu	2.43**	2.43**	2.60**	2.60**	.99	1.00
4. Coimbatore	Hindu	1.41	1.40	1.52	1.50	.98	.97
Malaysia							
5. Rural Malays	Hindu	3.12	3.28	4.39*	4.82*	1.08	1.06
6. Rural Malays	Chinese	10.90**	15.20**	27.40**	24.80**	1.13	.99
7. Market Malays	Hindu	8.10**	10.20**	8.13**	9.22**	1.05	1.03
8. Market Malays	Chinese	18.20**	22.20**	34.10**	35.90**	.99	.94
9. Urban Malays	Hindu	4.35**	4.78**	4.44*	4.95**	1.10	1.05
10. Urban Malays	Chinese	3.27**	4.83**	2.27	2.90	1.13	1.06
Thailand							
11. Malay-speakers	Buddhist	25.50**	107.80 ^d **	30.40**	— ^c	.09 ^d	— ^c
12. Thai-speakers	Buddhist	2.98**	3.06 ^d *	3.86*	3.87 ^d *	.79 ^d	.93 ^d
Philippines							
13. Zamboanga	Christian	1.78	1.13	1.50	.97	.84	.85
14. Zamboanga	Christian	1.63	1.57	1.45	1.41	.98	.99

*Significant at .05 level; **significant at .01 level

^a Estimates are net of age and parity. Age is controlled by introducing a third-order polynomial linear.Let $x = (\text{age} - \text{mean age})$, $Y = b_1 x + b_2 x^2 + b_3 x^3$. Parity is represented by a linear term for number of children and a contrast between those having less than two children and those having two or more.^b Freedom of movement scale described in text.^c SES (socioeconomic status) controls include controls on family income, household possessions, and wife's education.^d 1=can go all places without asking permission; 0 otherwise.^e Estimates are infinitely large.

freedom of movement influences the likelihood of being at a higher risk of an unwanted pregnancy.

While it has not been a focus of this analysis, some have argued that Muslim/non-Muslim differences reflect the disadvantaged socioeconomic position of Muslims. Data used here are strongest with respect to measurement of female autonomy but did include measures of education and a measure of income/wealth (presented in Table 2). These data did not show consistent Muslim disadvantage for the communities we studied. Consistent with this finding, the introduction of individual-level socioeconomic controls (in Table 6) does not attenuate the religious or ethnic effects. One can

see this by comparing coefficients in column 1 with those in column 3, or those in column 2 with those in column 4. If socioeconomic differentials were responsible for the observed fertility differences, then controls on these indicators (in columns 3 and 4) should have attenuated religious or ethnic differences shown in columns 1 and 2. No consistent attenuation is observed.

Summary and conclusion

In selected settings in four Asian countries, Muslim wives (compared to non-Muslim wives) usually have more children, are more likely to desire additional children, and are less likely to be using contraception when they desire no more children. Contrary to the primary hypothesis examined here, this greater pronatalist behavior/attitude cannot be accounted for by any general tendency of Muslim women to have less autonomy than non-Muslim women. We found very weak evidence for a link between religion or ethnicity and women's autonomy (also see Jejeebhoy and Sathar 2001) and no evidence at the individual level that women's autonomy was associated with fertility measures.

Despite credible theoretical arguments in support of the link between religion, women's status, and fertility, we argue that our empirical evidence to the contrary is strong. First, the data from the Survey on the Status of Women and Fertility contain direct measurement of multiple dimensions of women's autonomy. Regardless of whether one considers freedom of movement, economic power, or spousal coercion, results are consistent: autonomy differentials do not explain the greater pronatalist attitudes of the Muslim communities that we have studied. Second, our research design permitted multiple paired-comparisons across diverse settings. The literature contains many examples of comparisons of two settings. The weakness of such designs is that one is uncertain whether differences should be attributed to idiosyncratic explanations or to more general ones. With 14 comparisons we are in a stronger position to assess whether general explanations are appropriate. The fact that our comparisons span diverse sociocultural areas provides additional evidence that our findings can be generalized. Finally, our empirical work has examined the link of religion–autonomy–fertility at the appropriate conceptual level, the aggregate level. We have also considered the possibility that contextual-level effects could operate by shifting the distributions of important micro-level correlates of fertility.

While it has not been the focus of this analysis or the strength of the data we use, we have examined the hypothesis that greater pronatalist attitudes of Muslims result from greater poverty or other aspects of socioeconomic disadvantage. Controls on measures of women's education and on measures of household consumption have not attenuated the greater pronatalist attitudes of Muslim groups compared to non-Muslims.

These empirical results beg the question: in the settings we studied, what caused the greater pronatalist attitudes of Muslim compared to non-Muslim groups? Two general explanations (one focusing on women's autonomy, another on socioeconomic differences between groups) have found no support in our data. An alternative tack could focus on a set of separate explanations for these settings. As noted earlier, Jeffery and Jeffery (1997) offered a highly contextualized explanation for a single paired comparison of Muslims and Hindus in north India. No doubt differences in the social histories of these communities are relevant, but most of the communities we studied, like the pair studied by Jeffery and Jeffery, showed higher levels of childbearing among Muslims. Our future work will seek explanations for these similar outcomes despite unquestioned differences between these individual places. A possible explanation, suggested to us by various commentaries in the literature, links demographic difference to group identity and political disadvantage (e.g., Goldscheider and Uhlenberg 1969; Kreager 1986; Moghadam 1992; Siapno 1994; Basu 1997). To a great extent, religion is politics in this region of the world. Conflict between groups, drawn across explicitly religious lines, is central to South Asian cultures, and it is also relevant in Southeast Asia. The position of women and women's roles in reproduction occupy central positions in religious, popular, and political discourse. It may be conjectured that the Muslim community response to these concerns, especially when they are a minority, is a stronger emphasis on aspects of family life conducive to childbearing or opposition to particular forms of birth control, for example sterilization. Larger family sizes, greater demand for children, and higher risks of unwanted pregnancies could be the unintended outcomes of these group struggles.

Even if this speculation is correct, these religious and political effects favorable to higher levels of childbearing—which may lend conformity to the present experiences of Muslims in India, Malaysia, Thailand, and the Philippines—are nonetheless susceptible to change. The linking of largely ascribed characteristics to behavioral outcomes does not imply perpetual group differences. Earlier we noted that Islam is open to various interpretations regarding the roles of women and the acceptability of contraception. The dominant Muslim position on these issues is not immutable, nor is the saliency of such differences in political debates. The way these forces manifest themselves may vary sharply between countries or across regions within countries where Muslims are minority members and where they are not. But globalization of the economy and of communication also raises the possibility of strong pan-Islamic influences and responses.

We did not test this concluding speculation in our current work; rather it constitutes a post hoc explanation for the fertility differences that we could not explain by differences in women's autonomy or socioeconomic status.

Notes

This research was supported by Grants RO1 HD33757 and RO1 HD33791 from the National Institute of Child Health and Human Development to the East-West Center and the University of Pennsylvania. An earlier version of this article was presented at the 1998 Annual Meeting of the Population Association of America.

1 Muslims in certain regions of South and Southeast Asia have experienced recent and significant declines in fertility, e.g., in Indonesia. However, in the region overall there prevailed a general pattern for Muslims to display higher levels of fertility, a pattern that is still evident (see Kirk 1968; Weeks 1988).

2 In the SWAF data, respondents were asked why they did not use contraception. Muslims were generally more likely than Hindus to say "against religion," but it was not a response given routinely. In contrast, Knodel et al. (1999) report substantial evidence that Thai Muslims see contraception as "against their religion." Boonstra's (2001) overview is consistent with our claim.

3 Although direct injunctions, or *fatwas*, against sterilization have been issued by Islamic jurists, opposition is not unanimous. *Fatwas* state that use of modern methods resulting in permanent infertility contradict the will of Allah. Some jurists refer to prohibitions against infanticide and equate birth control with infanticide. On the other hand, sterilization operations for medical indications are rarely objected to because these operations are viewed as methods of treatment.

4 Specifically, the Jat community is politically and economically dominant, making low fertility (which prevents land fragmentation) and education of children preferred strategies.

5 Data, codebooks, and other descriptive material are available at «<http://www.pop.upenn.edu/swaf>»

6 The Chinese reported their religion primarily as Taoist, secondarily as Buddhist, with a smattering of Christians, other, and none. We have not subdivided this group because the relevant comparison is Chinese (an ethnic group) versus Muslim Malay (a religio-ethnic group).

7 Provinces in Thailand are many, hence small; communities in the same province are therefore close to one another. Differences between Muslims and (ethnic Thai) Buddhists in Songkhla Province are inevitably confounded by differences between rural and urban communities. There were roughly a dozen Buddhists in the rural Muslim community, and a similar number of Muslims in the urban Thai Buddhist community—not enough to sustain explicit comparisons. These small minorities have not been included in our calculations.

8 For formal analysis of differences across paired settings (i.e., in Table 5), the Thai data on freedom of movement cannot be usefully compared with data for the other three countries and are excluded.

9 A reduced set of these items was asked in paired settings 5 and 6. Thus, these data are excluded from calculations in Table 5, as described in note 8.

10 The second item was not asked in the Malaysian paired settings. Thus, these comparisons are not included in calculations shown in Table 5.

11 We restrict the sample to women who have been married at least ten years so that differences in fertility behavior could cumulate over time. At short durations the potential for substantial fertility differentials is small.

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